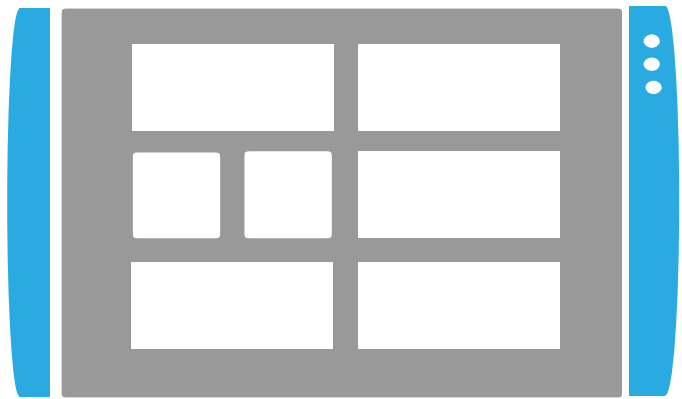
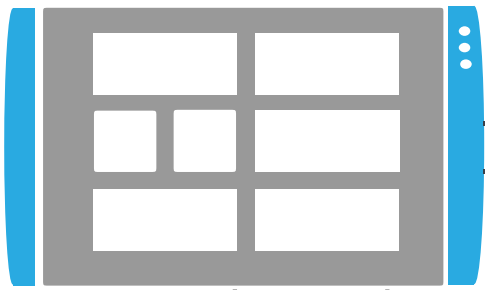




System Architecture of ESP32-Based IoT Smoke Monitoring System

Layer 1 – IoT Device



ESP32 Development Board
- Embedded Program
- Wi-Fi/ClientSecure (TLS)
- MQ-2 Smoke Sensor
- DHT22 Temperature & Humidity Sensor
- LED Indicator
- Buzzer
(Hardware: ESP32)

ESP32 Program Configuration



Publish Topic:
iot/sensor/data



Subscribe Topic:
iot/sensor/config



Protocol:
MQTT over TLS/SSL



QoS: 0



Port: 8883



Data Format: JSON



Outgoing JSON:
{
 "device_id": "ESP32-01",
 "smoke": true/false,
 "smoke_raw": integer,
 "temperature": float,
 "humidity": float
}



Incoming
Configuration JSON:
{
 "smokeThreshold": number,
 "detectInterval": number
}



Note:
Configuration updates
modify
- Smoke detection
 threshold
- Sensor monitoring
 interval

Subscribe iot/sensor/config
Protocol: MQTT/TLS
Port: 8883
Data: JSON
QoS 0

Publish iot/sensor/data
Protocol: MQTT/TLS
Port: 8883
Data: JSON
QoS 0

Layer 2 – Cloud MQTT Broker



Cloud: EMQX Cloud MQTT
Broker
Address:
v6612156.ala.asia-southeast1.emqxsl.com
Protocol:
MQTT over TLS/SSL
Port: 8883
Authentication: Username:
utem-iot-esp32,
utem-iot-nodered
(Technology: EMQX Cloud)

Protocol: MQTT/TLS
Port: 8883
Data: JSON



MQTT In
(MQTT v3.1 Legacy,
TLS Enabled, Port 8883)



Function:
Validate "device_id"



MongoDB Find
Collection:
device_metadata
Purpose:
Verify registered
device



Function:
Add "timestamp" &
Generate "local_time"



MongoDB insertOne
Collection:
sensor_data
Purpose:
Store time-series
sensor data

Layer 4 – Database



MongoDB Atlas
Database: utem-iot-assignment
Authentication:
Node-RED User
Privileges: readWrite
(Technology:
MongoDB Atlas)

Collection: device_metadata



Fields:
- device_id
- status
- location
- description

Collection: sensor_data (Time Series)

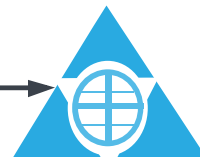


Fields:
- timestamp
- local_time
- device_id
- smoke
- smoke_raw
- temperature
- humidity

Layer 3 – Application Server



Ubuntu Cloud Server



Node-RED
Authenticated Login
(Technology: Node-RED)

Flow 1 – Sensor Data Processing



MQTT In
(MQTT v3.1 Legacy,
TLS Enabled, Port 8883)



Function:
Validate "device_id"



MongoDB Find
Collection:
device_metadata
Purpose:
Verify registered
device



Function:
Add "timestamp" &
Generate "local_time"



MongoDB insertOne
Collection:
sensor_data
Purpose:
Store time-series
sensor data

Flow 2 – Remote Configuration API



HTTP In
POST /iot/sensor/config



MQTT Out
Topic: iot/sensor/config
Retain: True



HTTP Response
200 OK



Purpose: Remote
update of
smokeThreshold and
detectInterval

Flow 3 – Device Registration API



HTTP In
POST /iot/sensor/register



MongoDB
insertOne
Collection:
device_metadata



HTTP Response
200 OK



Purpose: Register new
ESP32 device

Layer 5 – User Applications



MongoDB Charts
(Technology:
MongoDB Charts)



Postman Client
POST
http://20.196.222.43:1880/iot/sensor/register
JSON:
device_id, status,
location, description



Postman Client
POST
http://20.196.222.43:1880/iot/sensor/config
JSON:
smokeThreshold,
detectInterval